

DAY 5 NETWORK TESTING TOOL THEORY

Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

ANS:

What it does: It sends a tiny test message to YouTube to check if it's online and measures your connection speed (latency).

How to run it: Open your computer's Command Prompt (Windows) or Terminal (Mac) and type:
`ping [www.youtube.com](https://www.youtube.com)`

What average response time means: It shows how many milliseconds (`ms`) it took for your message to travel to YouTube and back.

- **10ms to 30ms:** Very fast and stable connection 🚀
- **Over 100ms:** Slow or lagging connection 🐢

Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.

ANS:

Command to run: Open your terminal and type `tracert`
`[www.flipkart.com](https://www.flipkart.com)` (Windows) or `traceroute`
`[www.flipkart.com](https://www.flipkart.com)` (Mac/Linux).

What it shows: Every router (hop) your data packet passes through on its way to Flipkart's servers.

Typical Hops Example:

1. **Hop 1:** Your home Wi-Fi router (`192.168.10.1`)
2. **Hops 2–3:** Your Internet Service Provider's (ISP) local and core routing switches
3. **Final Hop:** Flipkart's destination server

Use the nslookup tool to find the IP address of www.zomato.com and record the result.

ANS:

Command to run: Open your terminal and type `nslookup`
`[www.zomato.com](https://www.zomato.com)`.

What it does: Asks a DNS server to translate the domain name into its underlying server IP address.

Result: It queries the DNS and returns Zomato's server IP addresses (typically hosted on cloud load balancers or CDNs, such as `175.41.172.156` or similar regional server IPs).

Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

ANS:

Command to run: Type `ping 192.168.10.11` (using your phone's local IP address from our earlier subnet plan).

If the device is online: You will get quick, low-latency replies (often `<1 ms`) since it's on your local Wi-Fi.

If the device is offline (turned off or disconnected): You will see error messages like:

- **"Request timed out"** (the computer tried to reach it, but got no response).
- **"Destination Host Unreachable"** (your router couldn't find the device on the local network).

Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.
Constraint: Give a specific example for each tool based on your own test results above.

ANS:

Here is the troubleshooting summary report:

- **Basic Reachability (Website not loading):** Use `ping` to check if a server is online and responding, such as pinging `[www.youtube.com](https://www.youtube.com)` to check for packet loss.
- **Performance Bottlenecks (Slow connection):** Use `traceroute (tracert)` to isolate where latency occurs across individual router hops, like tracing the path to `[www.flipkart.com](https://www.flipkart.com)`.
- **Name Resolution (DNS issues):** Use `nslookup` to verify whether domain names correctly translate into server IP addresses, such as querying `[www.zomato.com](https://www.zomato.com)`.
- **Conclusion:** Combined, these three tools provide a complete end-to-end diagnostic workflow for any network issue.